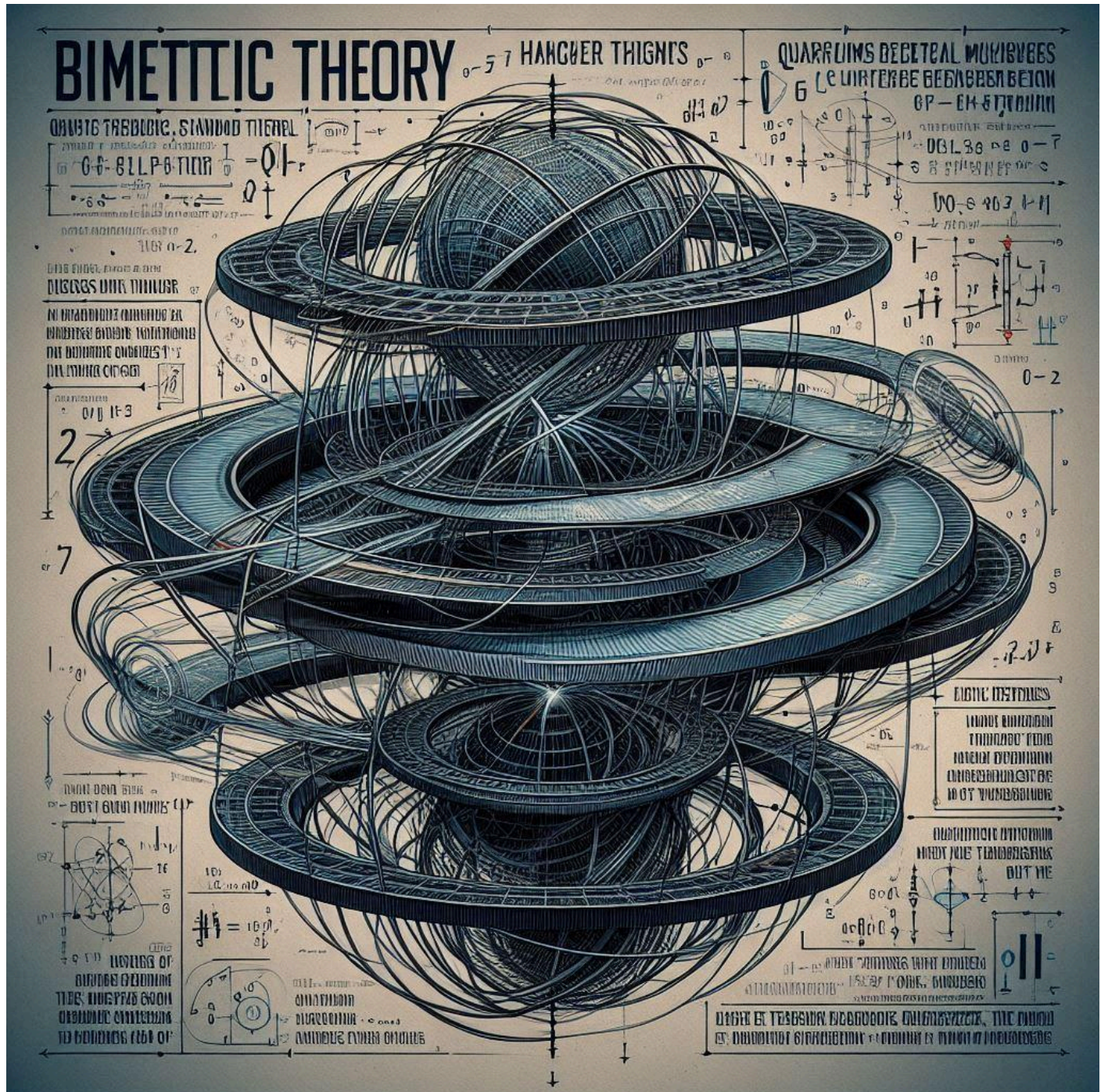


“Computational Physics, Penrose, Black Hole Metrics in Computational Physics. Fractals in Bimetric Ghost Theory in a Computational Framework”



Roberta Sparrow history(the philosophy of time travel)

- *Metrics*
- *Tensors*
- *Vectors*
- *Fractals in the theory of gravity or gravitational fields.*
- *This is a work focused on the translation of equations and metrics from books on gravity and black holes, merging them with ideas for the computational design of a plasma reactor. The next steps aim to design reactors or a new solar probe (like the Parker Solar Probe) and simulate it on paper.*



Roberta Sparrow's Story:

Roberta Sparrow was a science teacher and nun before becoming an almost "mythical" figure in Middlesex (the fictional town where Donnie lives).

In her youth, she wrote a book called "The Philosophy of Time Travel."

The book describes concepts such as:

- *Tangent Universes (unstable parallel worlds),*
- ***Objects trapped in time,***
- *Living and dead manipulators (characters who help or push key events),*
- *The Engine of Predestination (Artifact),*
- ***Wormholes*** *(wormholes as a gateway between realities).*
- *After writing the book, she seems to have had a revelation or traumatic experience that left her obsessed with death and time.*

Since then, she walks to her mailbox every day to check for a letter—as if waiting for a communication from the future or a sign that confirms her vision of time.

“Roberta Sparrow represents someone who discovered knowledge too vast for her time or her mind, and became trapped in her own vision of time. Her daily routine—walking to the mailbox, waiting—symbolizes persistent hope and the futility of absolute knowledge: she knows something is going to happen, but she can't directly intervene”

"The Philosophy of Time Travel" Guide (Structured Summary)

1. The Primary Universe

This is the "normal" universe, the one that exists without anomalies.

If everything follows its course, nothing strange should happen.

2. The Tangent Universe

Sometimes a short-lived parallel universe is generated by a tear in space-time (such as a gravitational anomaly).

It is unstable and must be closed before collapsing, or it will destroy the primary universe.

The airplane engine that falls on Donnie's house is an artifact that signals that the tangent universe has begun.

3. The Artifact

- *A metallic object falls into the tangent universe in an "inexplicable" way.*
- *It must be returned to the primary universe before it collapses.*
- *In the film, the artifact is the airplane engine that falls into Donnie's room.*

4. The Living Receiver

This person is chosen to guide the return of the Artifact.

They have special powers while the tangent universe is active:

- *Telekinesis,*
- *Foresight,*
- *Control of water and fire.*
- *In the film, Donnie Darko is the Living Receiver.*





- 1) In spacetime geometry (such as black holes or wormholes), the Riemann tensor describes how spacetime curves around masses or energies. They can be represented as points, paths and alterations of a curved manifold using tensors. Basic Tensor Representation (Idea)

-
1. The eye represents the point of maximum density of geodesic lines.
 2. The spiral represents the torsion or vorticity induced in space-time (this is literally what happens in a rotating Kerr-type black hole!).
 3. The tubes represent the flux lines of the vector field of possible futures (as Killing fields are visualized in relativity).

```
[Primary Spacetime ( $g_{\mu\nu}$ )]  
|  
[Spacetime Anomaly  $\rightarrow$  Local Positive Curvature (Riemann Tensor  $\neq 0$ )]  
|  
[Formation of a Tangent Universe with a Deformed Metric]  
|  
[Living Receiver Trajectory: Perturbed Geodesic Line]  
|  
[Vorticity Generation (Spiral), Projected Tensors (Future Tubes)]  
|  
[Wormhole Construction: Metric Deformation Between Regions]  
|  
[Final Solution: Stabilization or Collapse (Restoration of the Primary  $g_{\mu\nu}$ )]
```

To formalize it:

Space-time:

$(M, g_{\mu\nu})$

Curvature:

$R_{\sigma\mu\nu\rho}$ (Riemann tensor)

Living Receiver Trajectory:

$\gamma(t)$

Manipulated Forces:

$\nabla_{\gamma} T_{\mu}$ (manipulated tensor gradient)

Wormhole: Solution to $R_{\mu\nu} - (1/2) g_{\mu\nu} R = 8\pi T_{\mu\nu}$ with $T_{\mu\nu}$ of exotic matter.

```
--+
|                                     PRIMARY UNIVERSE (PU)
|
| Metric tensor:  $g(\mu\nu) \rightarrow$  Flat spacetime (Minkowski)
|
| Curvature Tensor:  $R(\rho\sigma\mu\nu) \approx 0$ 
|
| Living beings follow geodesics with no anomalies
|-----+
--+
|
|                                     ↓ [Anomaly detected at  $t = t_0$ ]
|-----+
--+
|                                     TANGENT UNIVERSE (TU)
|
| Metric tensor:  $g'(\mu\nu) \rightarrow$  Localized perturbation
|
| Artifact detected: Singular massive object (mass-energy anomaly)
|
| Curvature Tensor:  $R'(\rho\sigma\mu\nu) \neq 0$  (Nonzero local curvature)
|
| Wormhole begins formation (topological defect)
|-----+
--+
|
|                                     FIELD VECTORS OF MANIPULATED LIVING ( $F_{ML}$ )
|
|                                     |
|-----+
|                                     ↓ [Receiver trajectory deviation  $\gamma'(t)$ ]
|-----+
--+
|                                     LIVING RECEIVER  $\gamma'(t)$ 
|
| Attributes:
|
| • Future-tube perception ( $\partial\gamma/\partial t$  predicts future paths)
```

```

|   • Telekinetic anomalies (stress-energy deviation  $\Delta T(\mu\nu)$ )
|
|   • Visions through gravitational vorticity (spiral tensors)
|
|   • Wormhole manipulation potential
|
| Equation of motion:  $D^2\gamma'/dt^2 + R'(\gamma',\gamma')\gamma' = 0$  (geodesic deviation)
|
+-----+
--+
```

```

|
|                                     ↓ [Triggered Ensurance Trap Event]
|
+-----+
--+
```

FINAL WORMHOLE CREATION (WH)

```

|
| Artifact must be projected through wormhole to re-establish PU stability
|
| Tensor collapse:  $\Delta g(\mu\nu) \rightarrow 0$  at  $t = t_{\text{final}}$ 
|
| Energy condition violations allowed:  $T(\mu\nu)$  exotic
|
+-----+
--+
```

```

|
|                                     ↓ [Stabilization or collapse]
|
+-----+
--+
```

RESTORATION OF PRIMARY UNIVERSE

```

|
| Artifact's position reset at  $t = t_0$ 
|
| Tangent Universe erased from continuity
|
| Curvature Tensor:  $R(\rho\sigma\mu\nu) \rightarrow 0$  restored
|
+-----+
--+
```

$g(\mu\nu)$: Metric tensor (structure of space-time).

$R(\rho\sigma\mu\nu)$: Riemann tensor (measures curvature).

$\gamma'(t)$: Altered Living Receiver trajectory.

F_{ML} : manipulated force field.

$T(\mu\nu)$: Energy-momentum tensor (modified by anomalous events).

WH: Wormhole generated as a solution to extreme curvature.

A: Anomalous change in the structure of the metric or tensors.



```

+-----+
+-----+
|
|               PRIMARY UNIVERSE (PU)
|
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|
| Curvature Tensor:  $R(\rho\sigma\mu\nu) \approx 0$ 
|
| Living beings follow geodesics with no anomalies
|
+-----+
+-----+
|
|               [Anomaly detected at  $t = t_0$ ]
|
+-----+
+-----+
|
|               TANGENT UNIVERSE (TU)
|
| Metric tensor:  $g'(\mu\nu) \rightarrow$  Localized perturbation
|
| Artifact detected: Singular massive object (mass-energy anomaly)
|
| Curvature Tensor:  $R'(\rho\sigma\mu\nu) \neq 0$  (Nonzero local curvature)
|
| Wormhole begins formation (topological defect)
|
+-----+
+-----+
|
|
|
|               FIELD VECTORS OF MANIPULATED LIVING (F_ML)
|
|               [Emotional and causal forces driving Receiver's path]
|
|
|
|
+-----+
+-----+
|
|               [Receiver trajectory deviation  $\gamma'(t)$ ]
|
+-----+
+-----+

```

LIVING RECEIVER $\gamma'(t)$

Attributes:

- Future-tube perception ($\partial\gamma/\partial t$ predicts future paths)
- Telekinetic anomalies (stress-energy deviation $\Delta T(\mu\nu)$)
- Visions through gravitational vorticity (spiral tensors)
- Wormhole manipulation potential

Equation of motion: $D^2\gamma'/dt^2 + R'(\gamma',\gamma')\gamma' = 0$ (geodesic deviation)

↓ [Triggered Ensurance Trap Event]

FINAL WORMHOLE CREATION (WH)

Artifact must be projected through wormhole to re-establish PU stability

Tensor collapse: $\Delta g(\mu\nu) \rightarrow 0$ at $t = t_{\text{final}}$

Energy condition violations allowed: $T(\mu\nu)$ exotic

↓ [Stabilization or collapse]

RESTORATION OF PRIMARY UNIVERSE

Artifact's position reset at $t = t_0$

Tangent Universe erased from continuity

Curvature Tensor: $R(\rho\sigma\mu\nu) \rightarrow 0$ restored

What does "The Philosophy of Time Travel" say?

The book explains the rules that govern:

- The emergence of tangent universes.
- The existence of a "Living Receiver" (Donnie).
- The Manipulated (people who help or block the receiver).
- The role of Artifacts (such as the jet engine).
- The need to close the wormhole to prevent the destruction of the primary universe.

Roberta Sparrow(Physicist)



In theoretical physics, the graviton is the hypothetical particle that mediates gravity;

If graviton(s) existed, they would theoretically allow for:

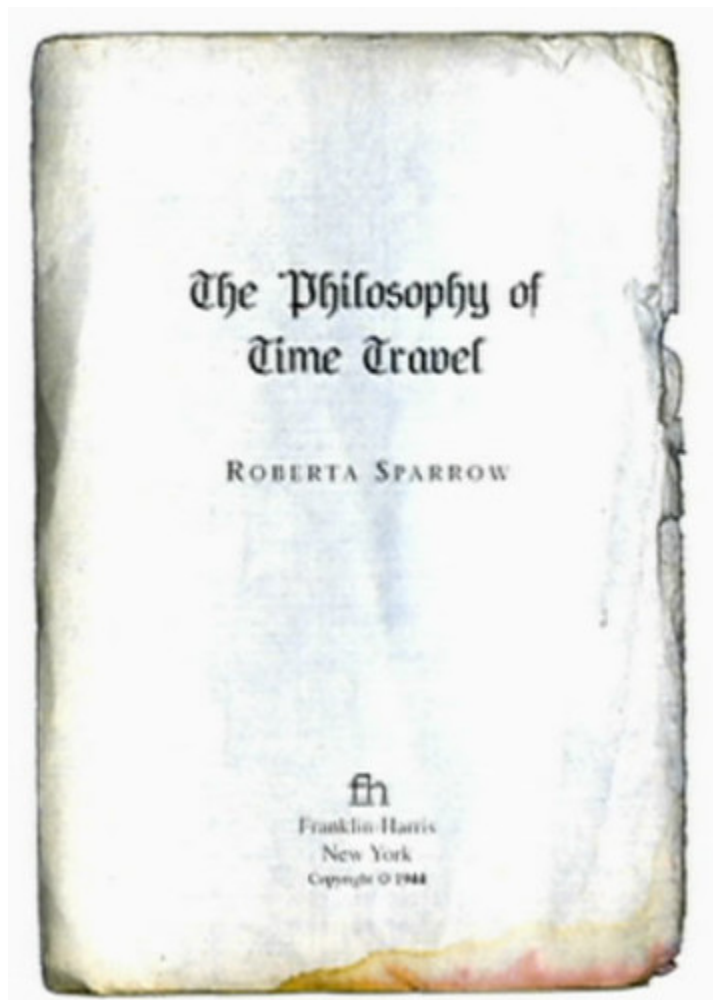
Wormholes. Connections between dimensions.

Time travel (according to some string or quantum gravity theories).

Roberta detects or senses anomalous gravity in the universe (just as a graviton detector would be able to sense invisible tensions in space-time).

That's why she waits in front of her mailbox:

"She knows" something will change, even if the rest of the world doesn't notice.



[[IMAGE; she waits in front of her mailbox]]





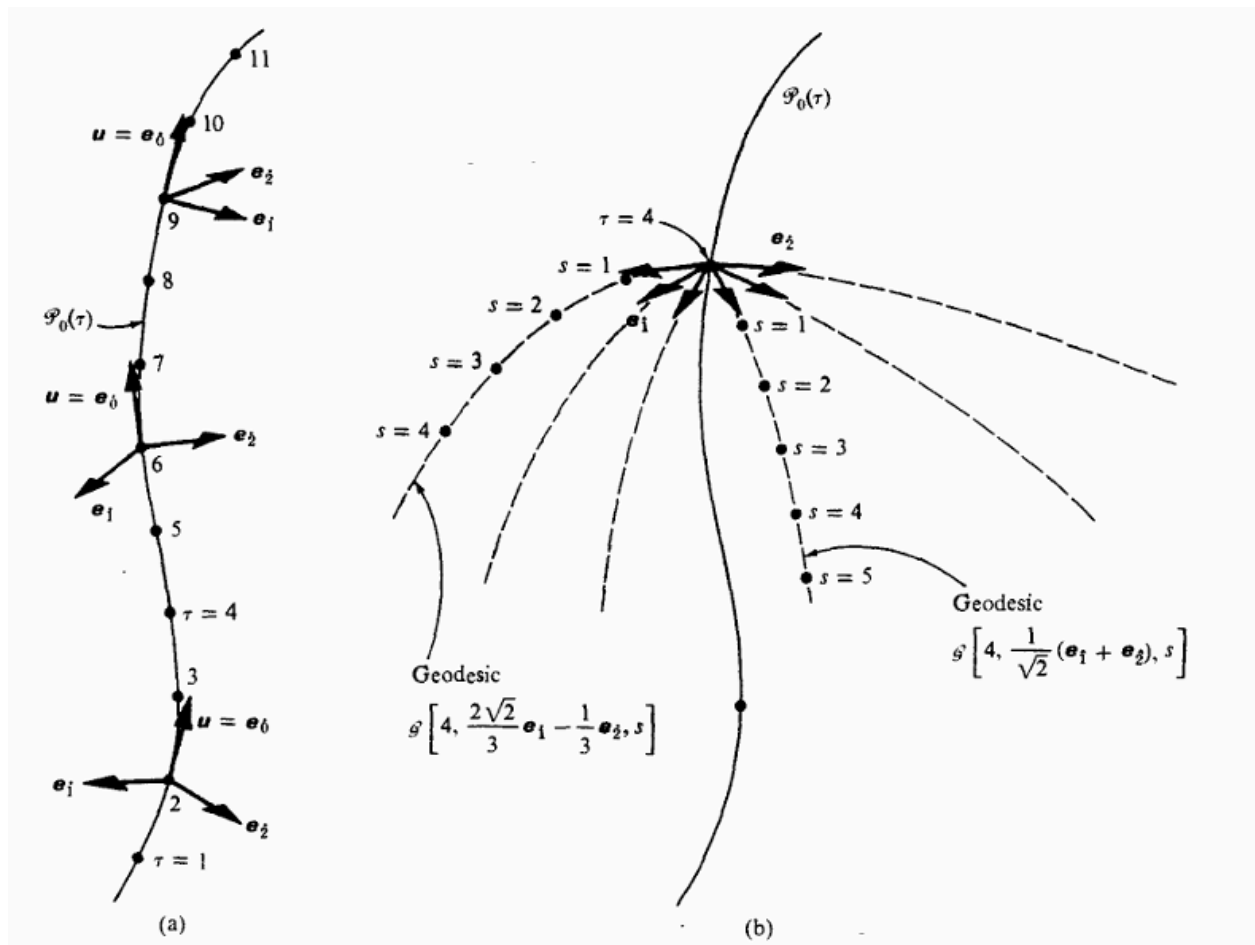
Important definitions:

A tokamak is a nuclear fusion device that controls super-hot plasma using toroidal magnetic fields.

Tokamak = *Toroidal'naya Kamera s Magnitnymi Katushkami* (Toroidal Chamber with Magnetic Coils in Russian). The plasma inside a tokamak is so energetic and dense that it begins to mimic extreme pressure and energy conditions, similar to those in the core of stars.

3. Relationship between Plasma and Gravitational Manipulation

Here's the interesting part:



If you manage to confine plasma in an extreme way (as in a tokamak or certain advanced reactors), we could create local energy-matter deformations. In theory, dense plasma + extreme gravitational fields = conditions similar to those needed to: Form a microscopic wormhole. Influence the curvature of space-time. On a practical level, we're still a long way from that. But the conceptual model is valid.

4. Space Gravity and Higher Topologies

"Space gravity is curved space or even some types of higher topologies..."

According to Einstein, gravity is not a "normal" force; it is a curvature of space-time caused by mass and energy.

Higher topologies are "strange" spaces with:

- Holes.
- Tunnels.
- Non-trivial connections (for example, a wormhole is an "Einstein-Rosen bridge").
- Manipulating these topologies involves:
- Having exotic matter (negative energy or non-standard energy-momentum tensors).
- Or generating energy densities so brutal that space-time itself bends like a black hole.

How to build an object with properties similar to a black hole or wormhole?

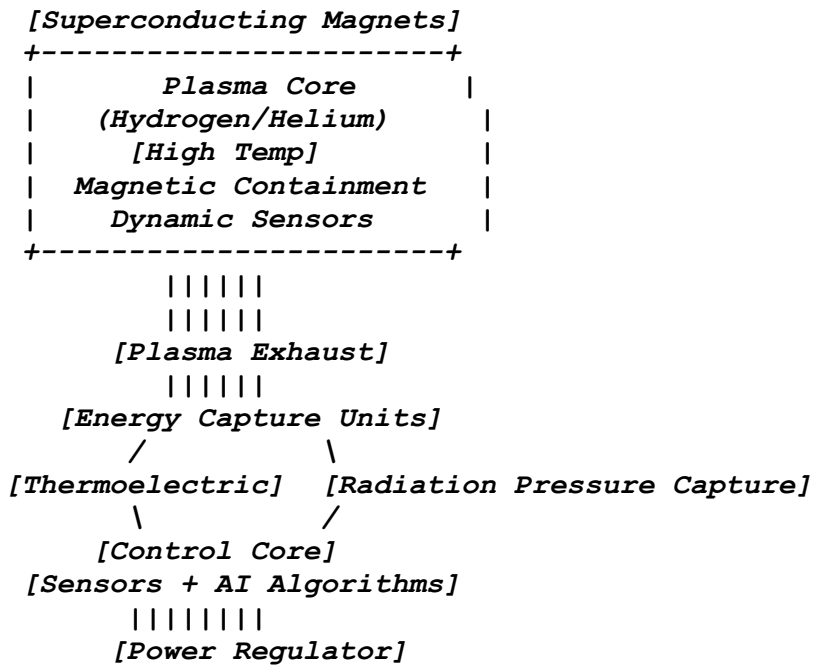
General theory:

Stage Description

1. Generate ultra-dense energy/matter fields (confined plasma, altered Higgs fields, etc.)
2. Obtain exotic matter (something with negative energy density relative to vacuum).
3. Configure the topology of space with appropriate Riemann tensors.
4. Maintain the open and stable structure through control fields (magnetic, gravitational, etc.).

Technical problems:

The energy required would be equivalent to that of a star. We still don't know how to make stable "exotic matter."



Stars: The closest stars we could use are red dwarfs or white dwarfs. Here are some examples:

Star Type	Distance	Useful Features
Proxima Centauri Red dwarf (M5.5Ve)	~4.24 light-years	Low mass, low radiation, stable. Usable plasma.
Barnard's Star Red dwarf (M4Ve)	~5.96 light-years	Very stable. Low solar activity.
Sirius B White dwarf	~8.6 light-years	Ultra-dense matter. High gravity. Dangerous but ideal if you can contain plasma.

What do we need to protect ourselves from radiation/solar flares?
Active shielding:

Huge magnetic fields, like a magnetic shield.

Controlled plasma shielding (like an artificial "plasma bubble").

Material shielding:

Layers of ultra-dense materials like liquid hydrogen or metallic foam.

Physical layers in the capsule:

Such as a shell of super-dense cells or topological materials (like hyper-resistant 3D graphene).

The human capsule would need to be buried within at least 5-10 meters

of equivalent shielding to withstand a nearby coronal mass ejection.

Models:

```
+-----+
|           Plasma Control Reactor Core - PCGCR           |
+-----+
| 1. Plasma Injection from Stellar Source                  |
| 2. Magnetic Containment Coils Activated (Toroidal)      |
| 3. Plasma Density and Temperature Sensors Initialized   |
| 4. Stabilization Feedback Loop (Electromagnetic Pulse)  |
| 5. Energy Extraction Interface (Thermoelectric Modules) |
+-----+

# Modelo de curvatura artificial mínima
import numpy as np

# Tensor de Riemann simulado en formato 4x4x4x4
riemann_tensor = np.zeros((4,4,4,4))

# Suponemos que el plasma genera curvatura en las direcciones espaciales
# Añadimos perturbaciones simuladas
riemann_tensor[0,1,0,1] = 1e-15 # Tiempo-espacio x
riemann_tensor[0,2,0,2] = 1e-15 # Tiempo-espacio y
riemann_tensor[0,3,0,3] = 1e-15 # Tiempo-espacio z

print("Riemann Tensor Components (simulated):")
print(riemann_tensor)

# Plasma Control Reactor Core - Simulation

# Step 1: Initialize plasma properties
plasma_temperature = 1e7 # Kelvin
plasma_density = 1e20    # particles/m^3

# Step 2: Magnetic Field Initialization
magnetic_field_strength = 5.0 # Tesla

# Step 3: Feedback control loop to maintain stability
while plasma_temperature > 5e6:
    # Sensor reads
    current_density = plasma_density
    current_temperature = plasma_temperature

    # Magnetic field adjustment based on sensors
    if current_temperature > 1e7:
```

```

        magnetic_field_strength += 0.5 # Strengthen containment
    elif current_temperature < 6e6:
        magnetic_field_strength -= 0.2 # Relax containment

# Simulate plasma cooling and stabilization
plasma_temperature *= 0.99 # Small cooling per cycle
plasma_density *= 0.999 # Minor particle loss

# Energy extraction simulation
extracted_energy = plasma_temperature * plasma_density * 1e-12 #
arbitrary units

# Display reactor status (compact txt print)
print(f"Temp: {plasma_temperature:.2e} K | Density:
{plasma_density:.2e} | Field: {magnetic_field_strength:.2f} T | Energy
Extracted: {extracted_energy:.2e}")

print("Plasma stabilized. Energy extraction at steady rate.")

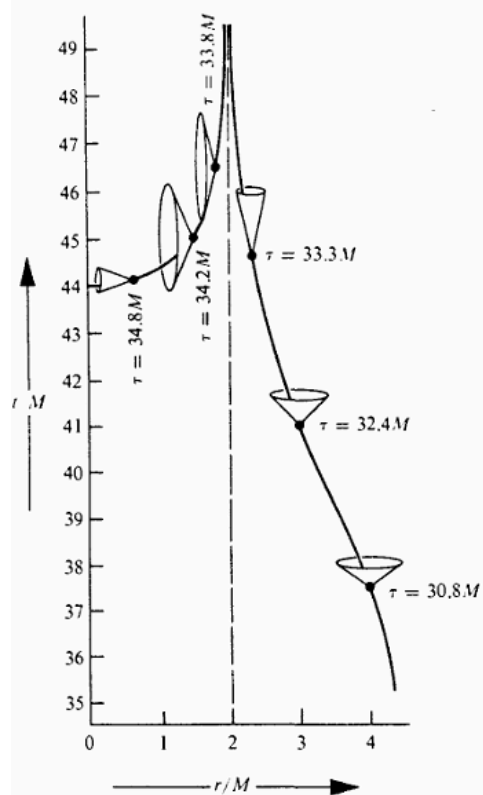
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2nd model: [Superconducting Magnets]

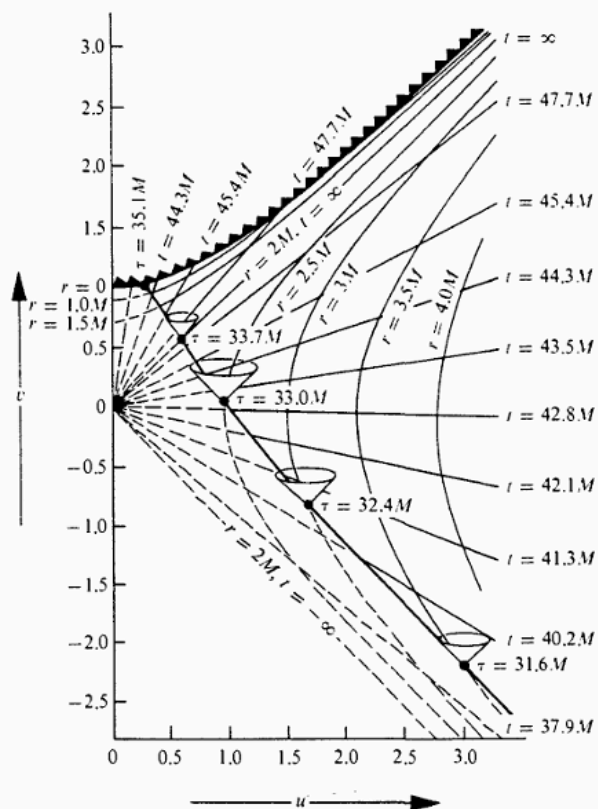
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+-----+
|      Plasma Core      |
| (Hydrogen/Helium)     |
| [High Temp]           |
| Magnetic Containment  |
| Dynamic Sensors       |
+-----+
      |||||
      |||||
    [Plasma Exhaust]
      |||||
    [Energy Capture Units]
      /      \
[Thermoelectric] [Radiation Pressure Capture]
      \      /
    [Control Core]
[Sensors + AI Algorithms]
      |||||
    [Power Regulator]

```

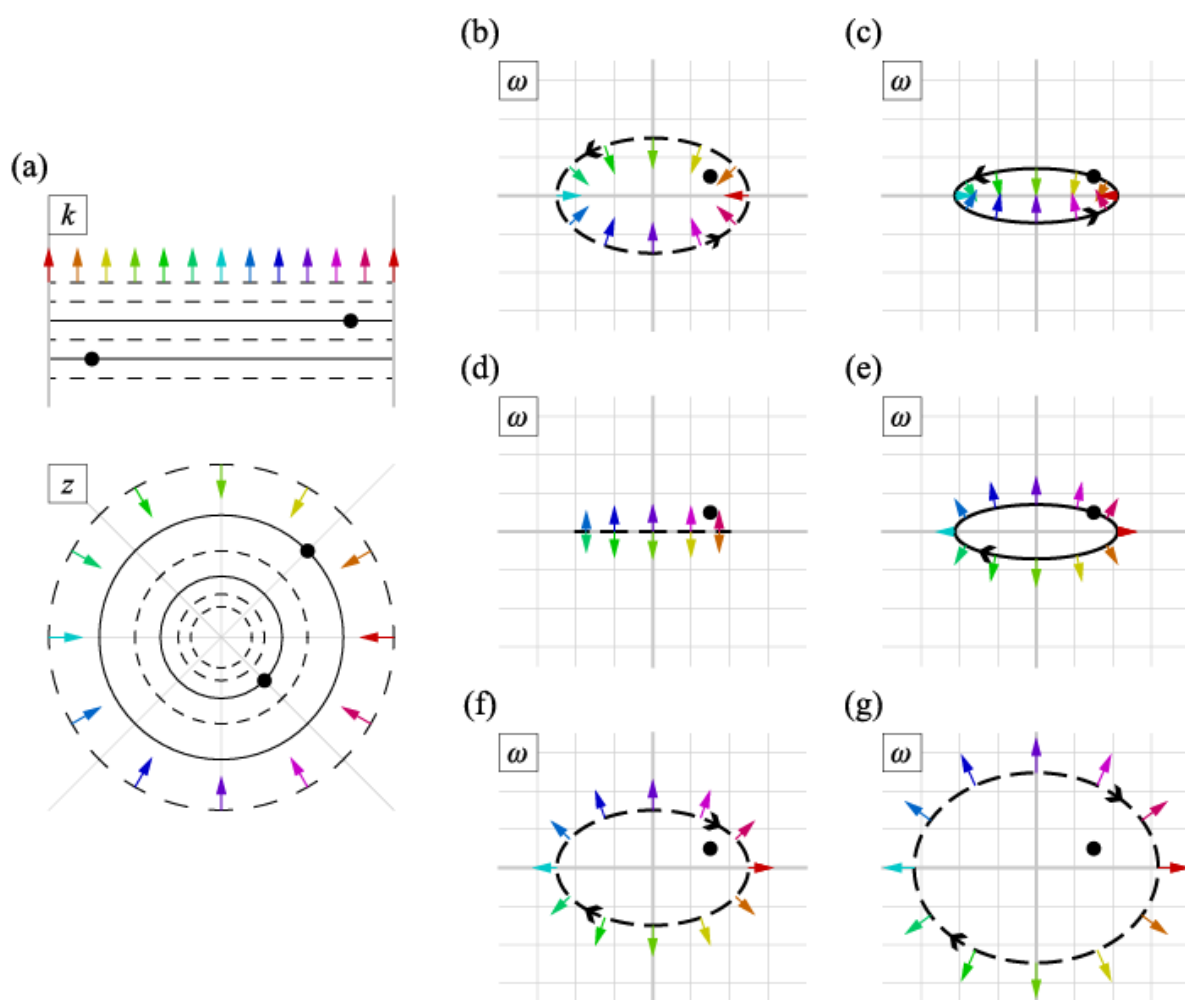



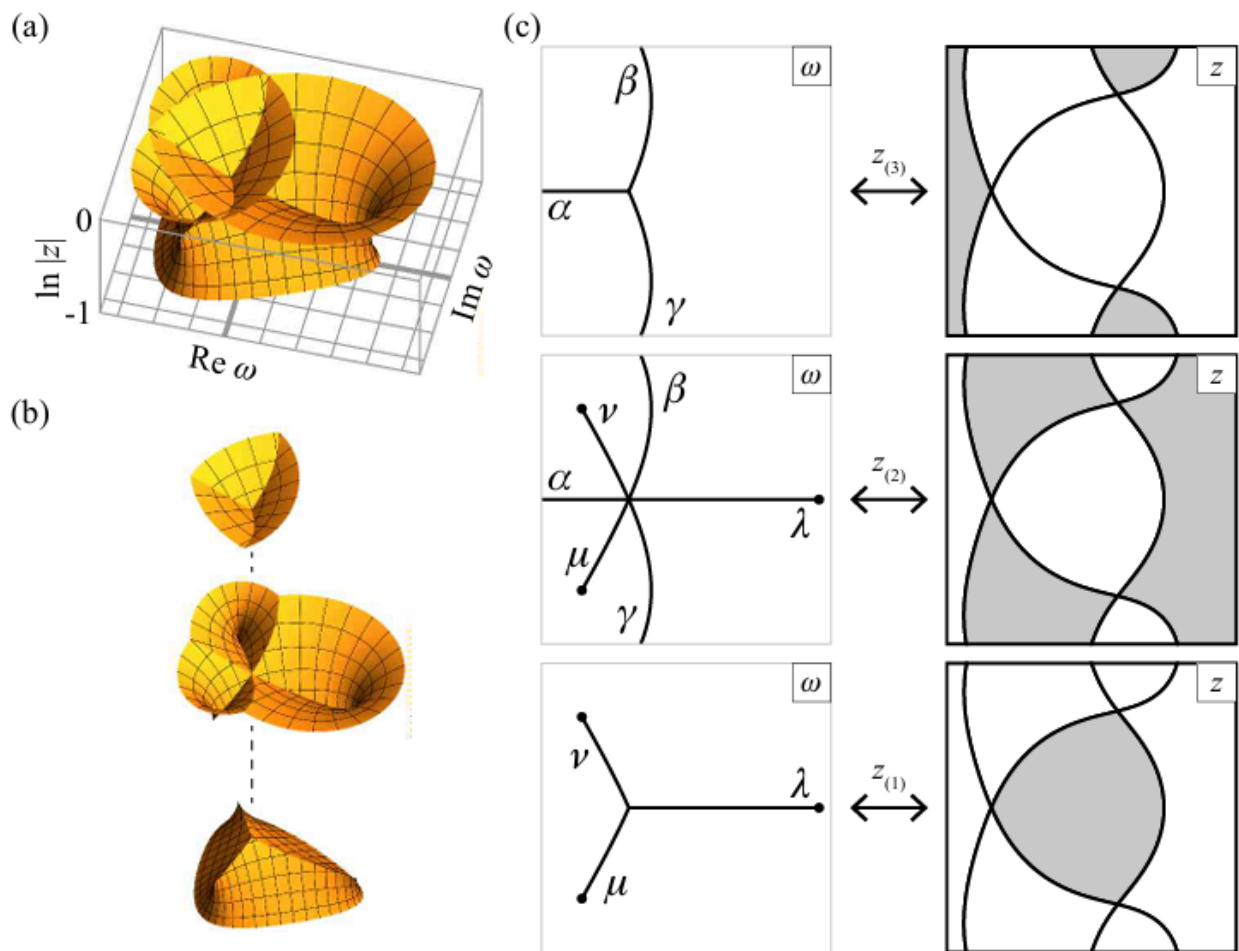
(a) Schwarzschild



(b) Kruskal-Szekeres

Figure 15





$$\frac{d}{d\tau} \begin{bmatrix} \psi_{A,n} \\ \psi_{B,n} \end{bmatrix} = -i \sum_m \begin{bmatrix} t_{AA,m} & t_{AB,m} \\ t_{BA,m} & t_{BB,m} \end{bmatrix} \begin{bmatrix} \psi_{A,n+m} \\ \psi_{B,n+m} \end{bmatrix}.$$